# Exploratory Data Analysis for Fraudulent Job Posting Detection

Employment Scam Aegean Dataset - EMSCAD

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This report presents dataset quality, class imbalance, missing-value behaviour, textual patterns, categorical risk factors, and data-leakage findings.

# Executive Summary

- The dataset contains 17,880 job advertisements.

- Only 866 advertisements are fraudulent, producing strong class imbalance.

- Accuracy alone is unsuitable because an all-genuine model reaches about 95.16 percent accuracy.

- Missing company profiles and absent company logos are strong fraud indicators.

- Fraudulent descriptions and requirements are generally shorter.

- Work-from-home and no-experience language show high fraud risk.

- Repeated advertisements and text templates create serious leakage risk.

- The primary evaluation should use a group-aware split and independent real-world testing.

## Important methodological decision

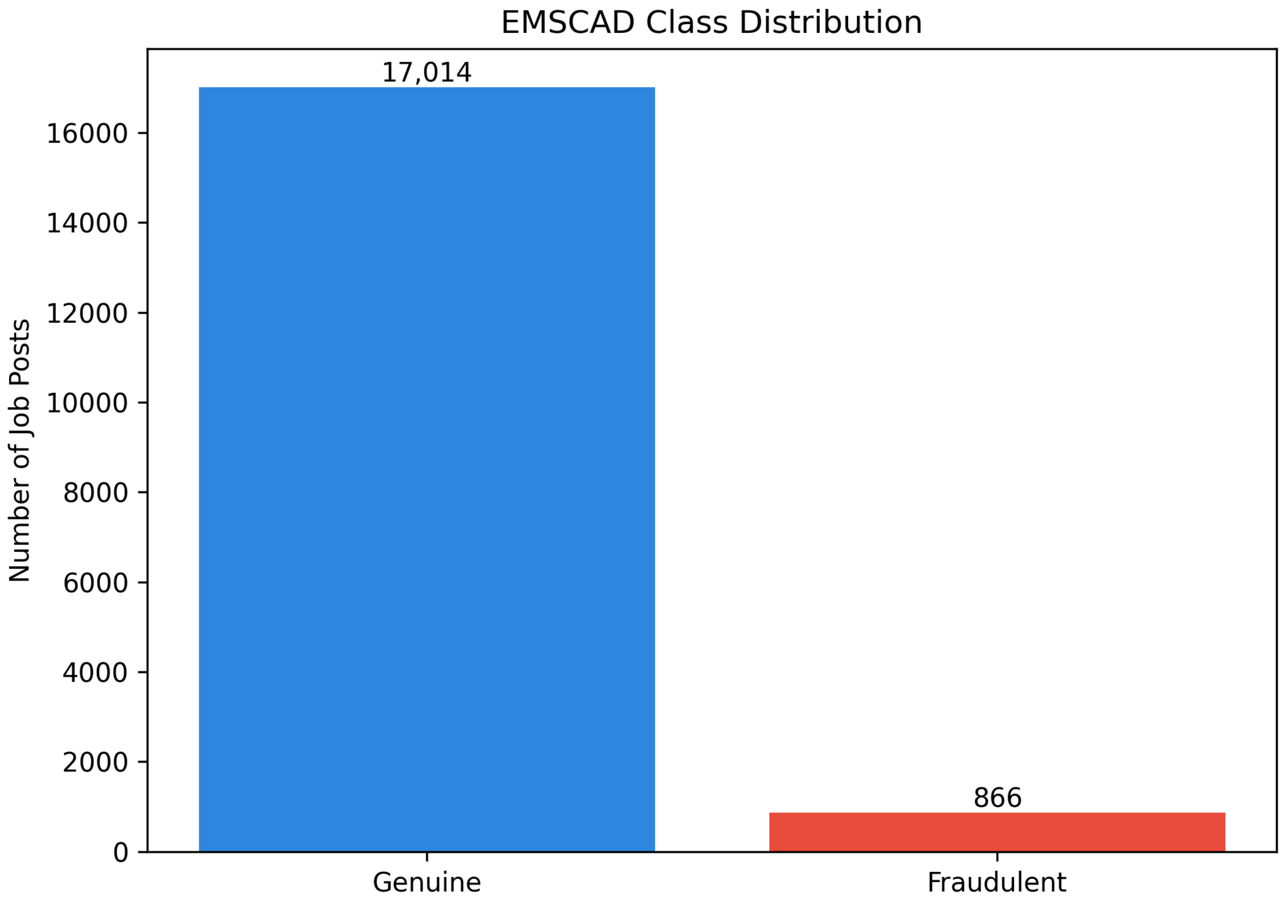
For comparison with existing papers, a standard stratified evaluation may be reported separately. However, the main research claim should use leakage-safe group-aware validation so that repeated advertisements and templates cannot appear in both training and testing data.

# 1. Dataset Overview

| Metric | Value |
|---|---|
| Total rows | 17880.00 |
| Total columns | 18.00 |
| Genuine posts | 17014.00 |
| Fraudulent posts | 866.00 |
| Fraud percentage | 4.84 |
| Exact duplicate rows | 0.00 |
| Extra content duplicate copies | 281.00 |

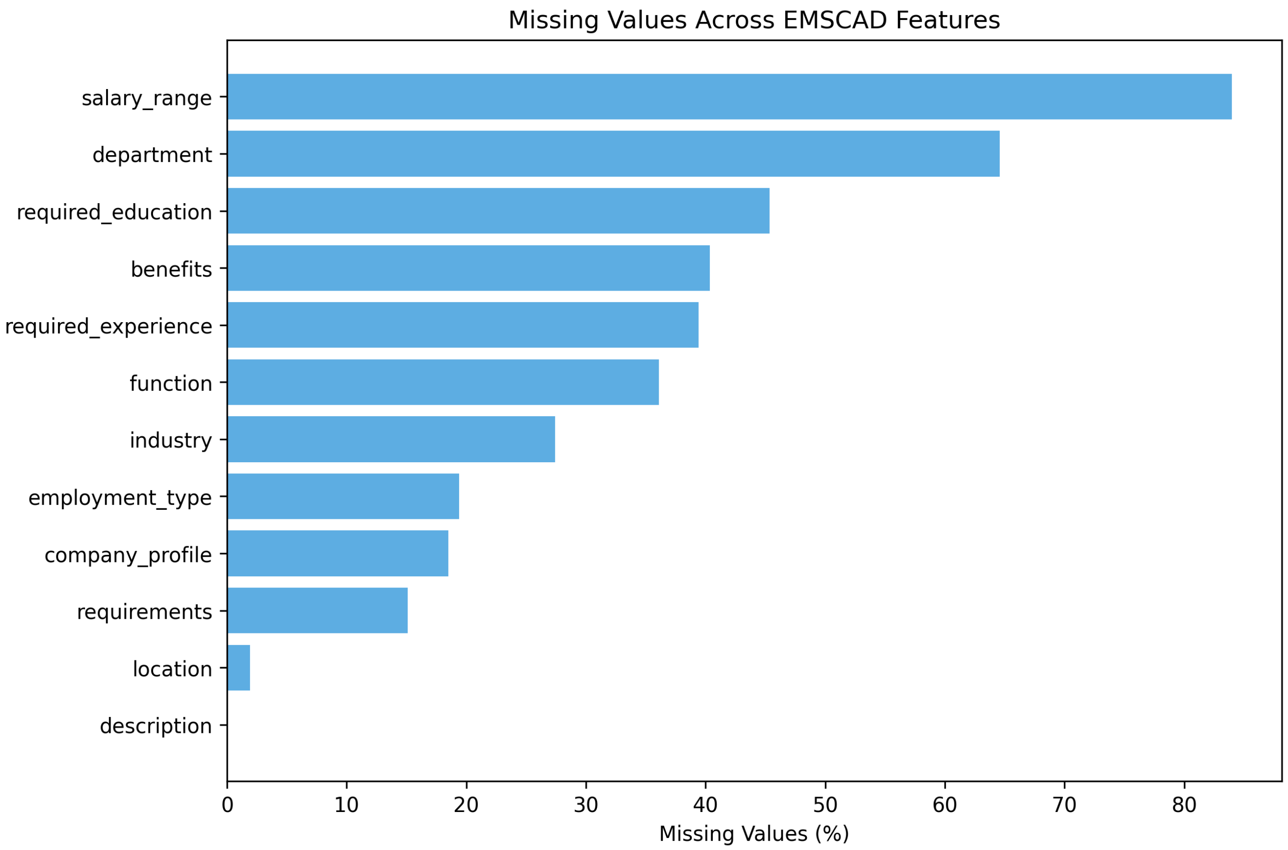
The fraudulent class represents approximately 4.84 percent of all records. Therefore, precision, recall, F1-score, PR-AUC and MCC should be treated as primary evaluation metrics.

## 2. Target-Class Distribution



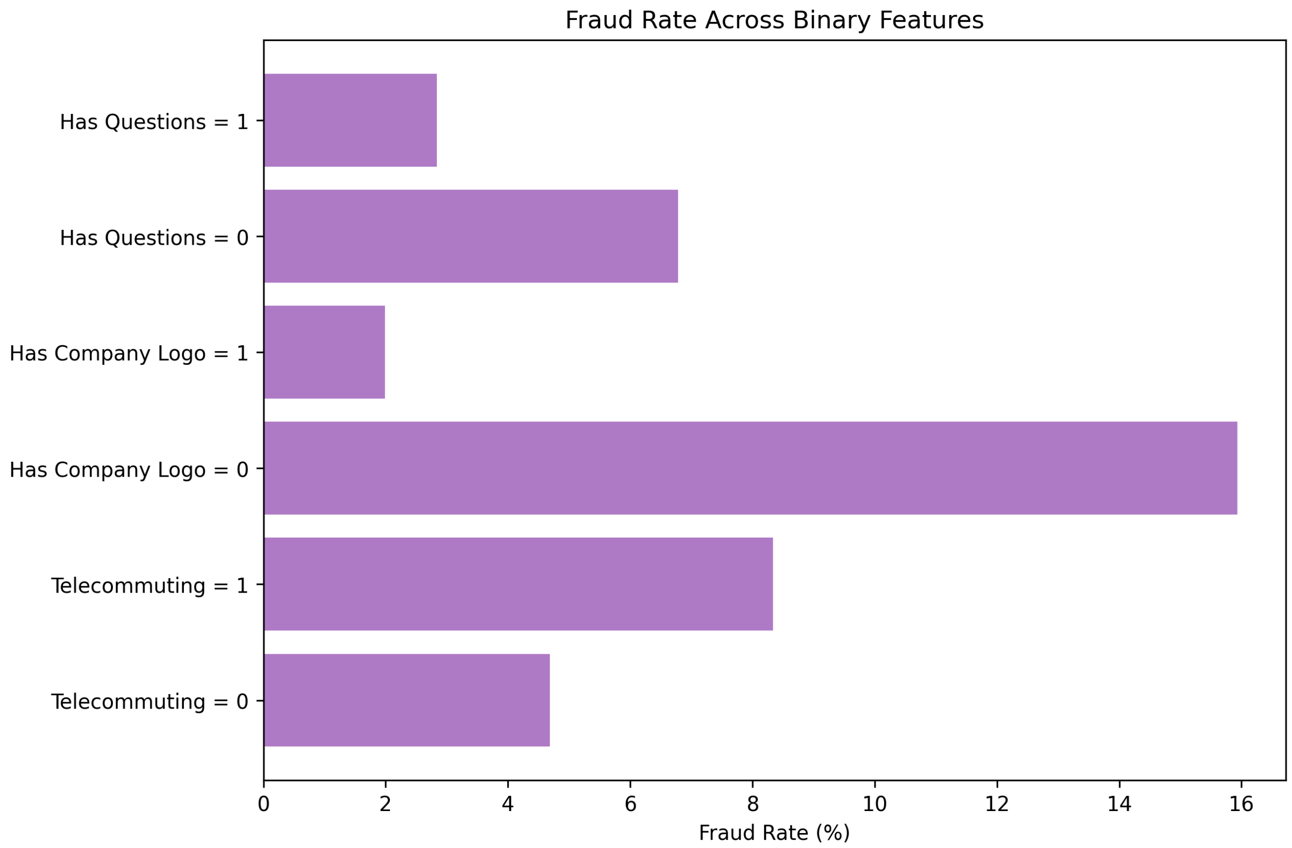

*The dataset is highly imbalanced. A high accuracy score does not automatically mean that fraudulent jobs are detected.*

# 3. Missing-Value Distribution



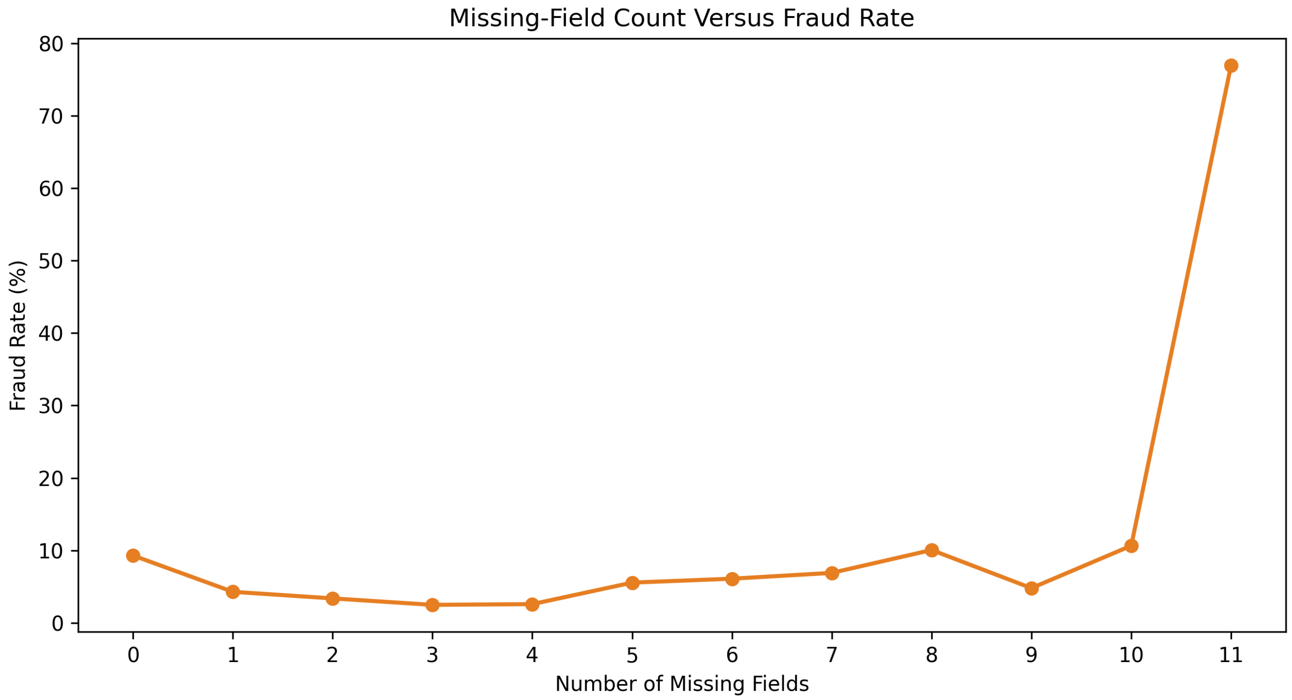

*Salary range and department have the largest missing-value percentages. Missing status should be preserved as a model feature.*

# 4. Binary Metadata Signals



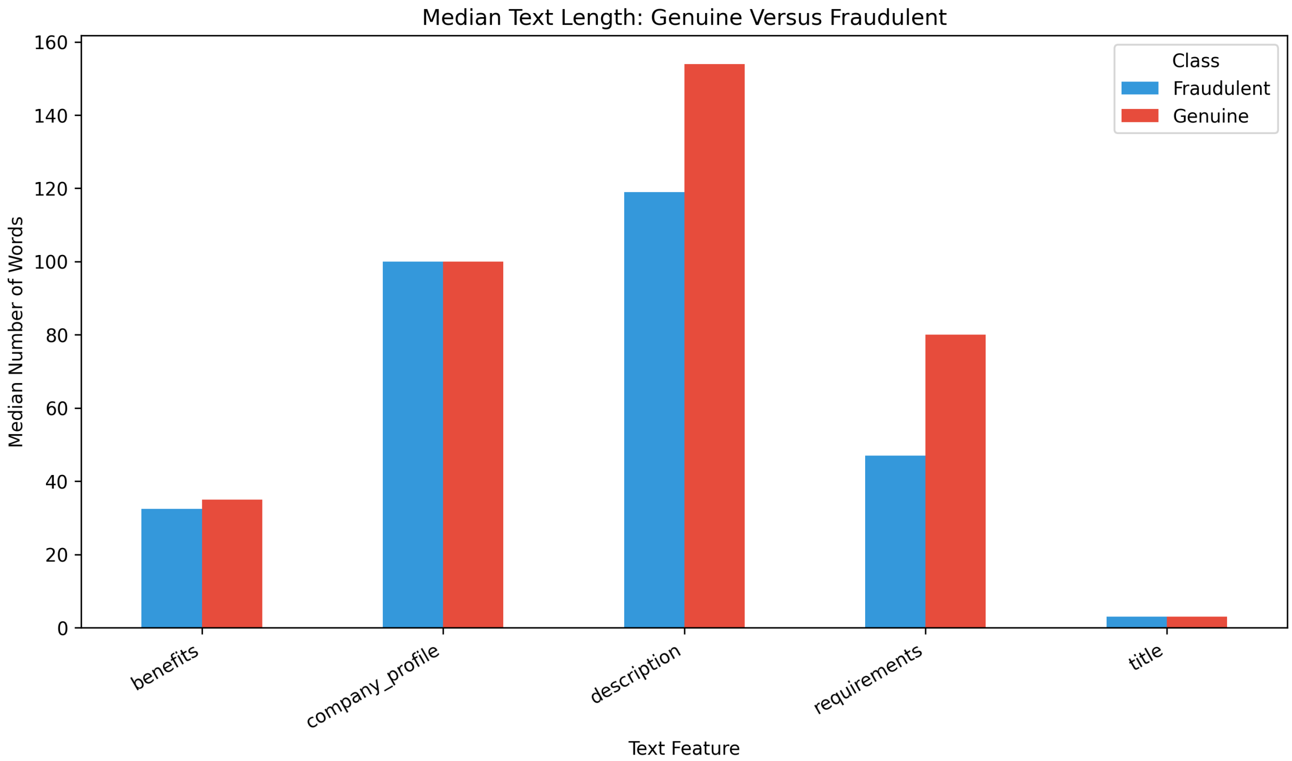

*Company-logo availability, screening questions and telecommuting status provide useful metadata signals.*

# 5. Missing-Field Count



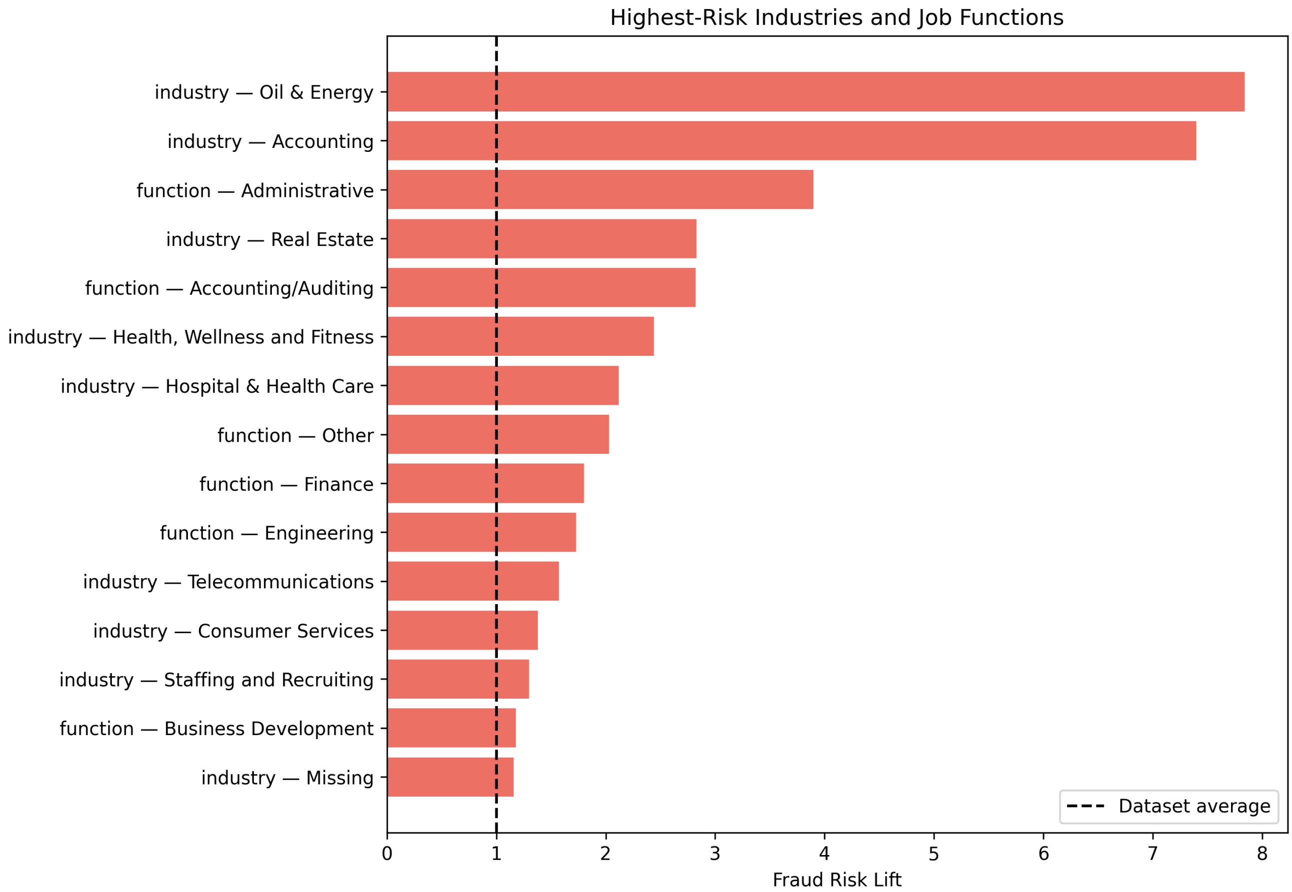

*The relationship between total missing fields and fraud is non-linear, supporting non-linear feature interactions.*

# 6. Text-Length Comparison



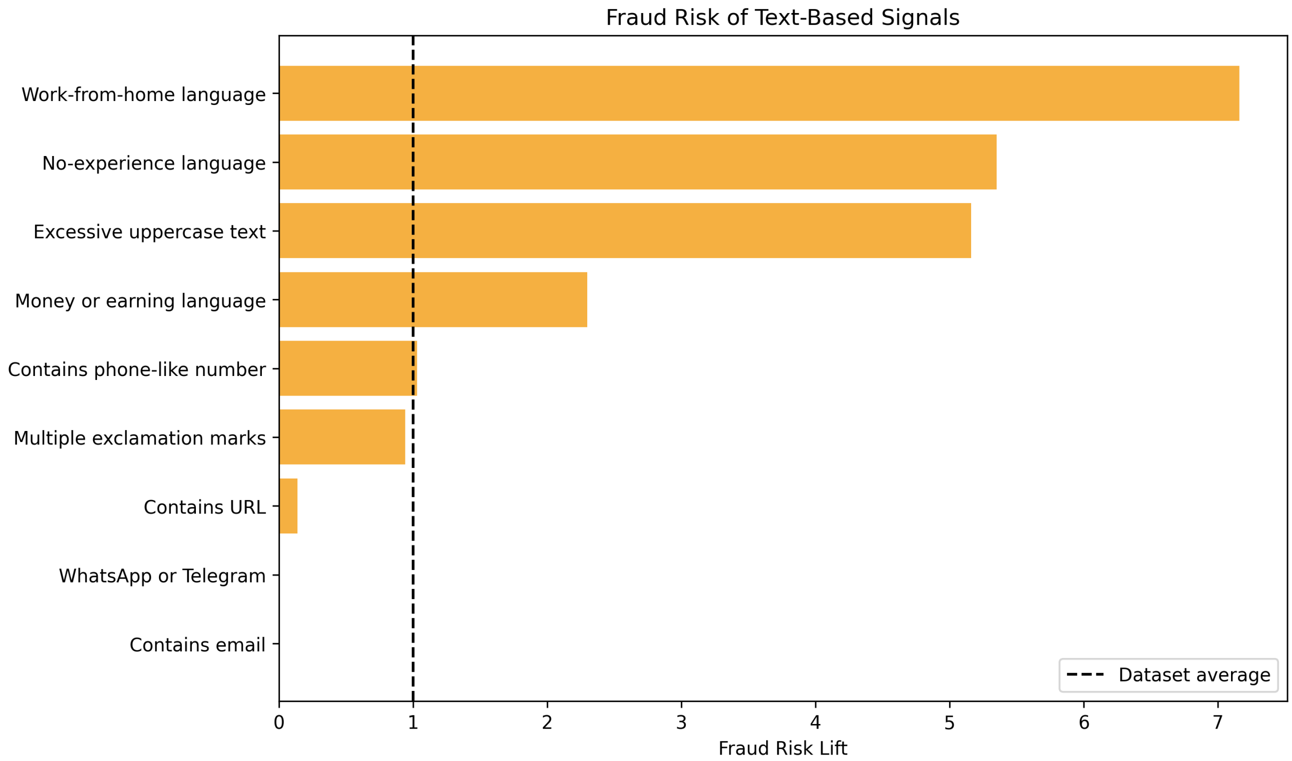

*Fraudulent descriptions and requirements are generally shorter than genuine advertisements.*

# 7. Industry and Function Risk



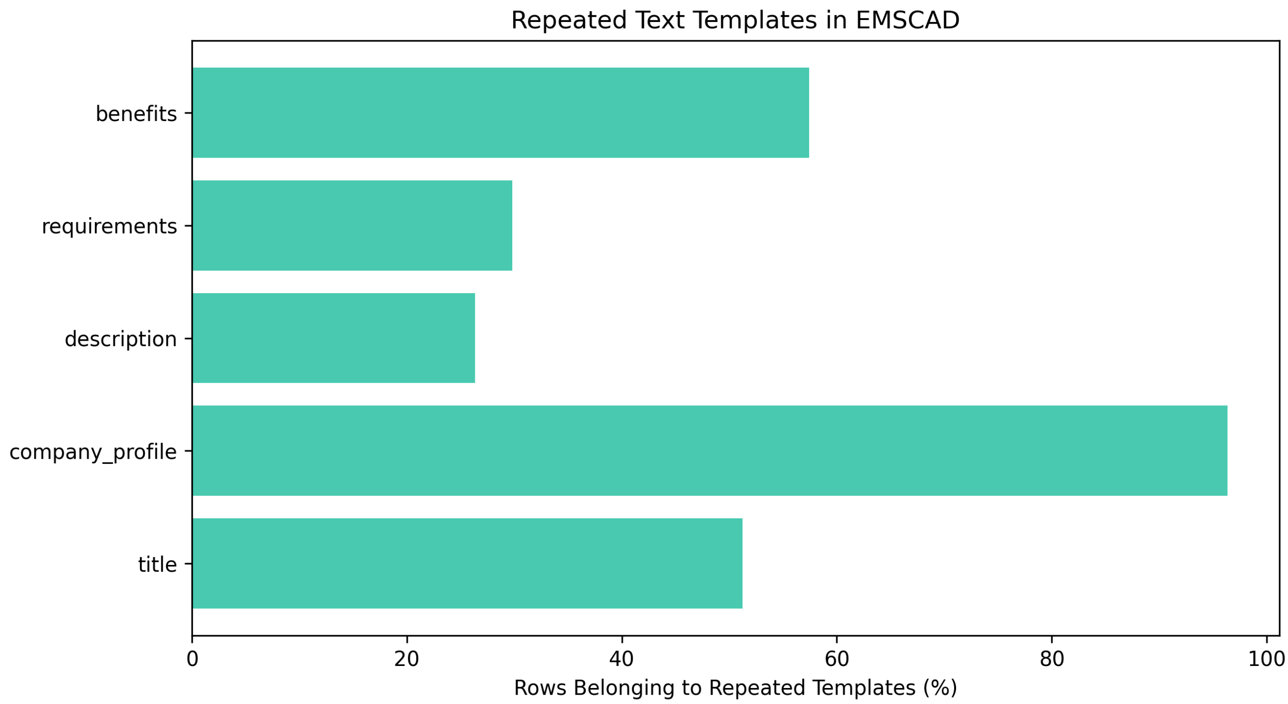

*Some industries and job functions show much higher fraud rates, but these patterns may change in future real-world data.*

# 8. Text-Based Risk Signals




*Work-from-home, no-experience and earning-related language show strong fraud associations.*

# 9. Repeated Text Templates




*Large numbers of advertisements reuse text templates. Random splitting can therefore produce over-optimistic results.*

# 10. Missing-Value Details

| Feature | Missing count | Missing percentage |
|---|---|---|
| salary_range | 15,012 | 83.96 |
| department | 11,547 | 64.58 |
| required_education | 8,105 | 45.33 |
| benefits | 7,212 | 40.34 |
| required_experience | 7,050 | 39.43 |
| function | 6,455 | 36.10 |
| industry | 4,903 | 27.42 |
| employment_type | 3,471 | 19.41 |
| company_profile | 3,308 | 18.50 |
| requirements | 2,696 | 15.08 |
| location | 346 | 1.94 |
| description | 1 | 0.01 |

# 11. Terms Associated With Fraudulent Posts

| Term | Fraud documents (%) | Genuine documents (%) | Log-odds score |
|---|---|---|---|
| leveraging career | 9.01 | 0.00 | 8.13 |
| aptitude staffing | 7.04 | 0.00 | 7.86 |
| services town | 6.47 | 0.00 | 7.77 |
| represented candidates | 6.47 | 0.00 | 7.77 |
| participate referral | 6.47 | 0.00 | 7.77 |
| following perks | 6.47 | 0.00 | 7.77 |
| encouraged participate | 6.47 | 0.00 | 7.77 |
| expert negotiations | 6.47 | 0.00 | 7.77 |
| field employ | 6.00 | 0.00 | 7.69 |
| url_0fa3f7c5e23a16de16a841e368006cae916884407d90b154dfef3976483a71ae information | 6.00 | 0.00 | 7.69 |
| recovery petroleum | 6.00 | 0.00 | 7.69 |
| petroleum field | 6.00 | 0.00 | 7.69 |
| people values | 6.00 | 0.00 | 7.69 |
| production maximize | 6.00 | 0.00 | 7.69 |
| provider products | 6.00 | 0.00 | 7.69 |

Several terms appear to represent specific organizations or repeated fraud campaigns. They must not be treated as universal fraud rules.

# 12. Terms Associated With Genuine Posts

| Term | Fraud documents (%) | Genuine documents (%) | Log-odds score |
|---|---|---|---|
| university degree | 0.00 | 5.82 | -4.67 |
| asia | 0.00 | 5.57 | -4.63 |
| 1500 | 0.00 | 5.46 | -4.61 |
| medium large | 0.00 | 4.41 | -4.38 |
| amp secure | 0.00 | 4.42 | -4.38 |
| teaching experience | 0.00 | 4.36 | -4.37 |
| tefl | 0.00 | 4.32 | -4.36 |
| tesol | 0.00 | 4.30 | -4.36 |
| help teachers | 0.00 | 4.30 | -4.36 |
| student loans | 0.00 | 4.28 | -4.35 |
| teachers safe | 0.00 | 4.27 | -4.35 |
| secure jobs | 0.00 | 4.27 | -4.35 |
| tefl tesol | 0.00 | 4.29 | -4.35 |
| jobs abroad | 0.00 | 4.27 | -4.35 |
| celta teaching | 0.00 | 4.27 | -4.35 |

# 13. Text-Template Leakage Audit

| Feature | Non-empty | Unique | Repeated groups | Rows repeated | Largest group | Fraud only | Genuine only | Mixed labels |
|---|---|---|---|---|---|---|---|---|
| title | 17,880 | 10,533 | 1,810 | 9,157 | 406 | 100 | 1,630 | 80 |
| company_profile | 14,572 | 1,706 | 1,174 | 14,040 | 726 | 36 | 1,138 | 0 |
| description | 17,878 | 14,442 | 1,269 | 4,705 | 417 | 94 | 1,174 | 1 |
| requirements | 15,178 | 11,768 | 1,113 | 4,523 | 410 | 96 | 1,011 | 6 |
| benefits | 10,646 | 5,806 | 1,275 | 6,115 | 726 | 58 | 1,207 | 10 |

Repeated and class-specific templates can allow a model to memorize campaigns instead of learning general fraud behaviour. Template or campaign groups must remain within a single data split.

# 14. Final EDA Conclusions

- Keep the complete 17,880-row EMSCAD dataset for paper comparison.

- Create missing-value indicators instead of blindly deleting incomplete columns.

- Process long text, categorical metadata and engineered risk signals through separate model branches.

- Parse salary and location into structured features.

- Fit vectorizers, encoders, imputers and resampling methods using training folds only.

- Prevent duplicate advertisements and repeated templates from crossing data splits.

- Use PR-AUC, fraud recall, fraud precision, F1-score, MCC, ROC-AUC and accuracy.

- Tune the classification threshold using validation data only.

- Perform ablation studies to prove which proposed component adds value.

- Test the final model using independent and recent real-world advertisements.

## EDA Status: Completed